

Supplementary Methods: mathematical specification

Calendar, state boundary and developmental clocks

The daily forcing comprises mean temperature T_d and maximum temperature $T_d^{\max}$ in °C, mean relative humidity R_d in percent, and precipitation P_d in mm. The crop indicator c_d equals one from the supplied sowing date through the supplied crop-end date, inclusive. Regional seasons end at the first predicted BBCH85 crossing. The state indexed by d is the state at the end of day d ; initial developmental clocks are zero. Weather outside the crop window advances the calendar while freezing developmental clocks, tissue and local source potential. The forcing is consecutive, and a fresh simulation includes the sowing date. Accepted daily mean temperatures are at most 40 °C, the upper support of the retained thermal response.

Let $\text{clip}(x, a, b) = \min\{b, \max(a, x)\}$. Daily effective thermal accumulation is

$$\begin{aligned} g_d &= c_d \begin{cases} \text{clip}(T_d, 0, 20), & T_d \leq 30, \\ 20 - 2(T_d - 30), & 30 < T_d \leq 40, \end{cases} \\ G_d &= G_{d-1} + g_d. \end{aligned} \quad (\text{S1})$$

For latitude ϕ in radians and calendar day of year j_d , the retained astronomical and photoperiod functions are

$$\begin{aligned} \delta_d &= 0.4093 \sin\{2\pi(j_d - 81)/365\}, \\ \ell_d &= \frac{24}{\pi} \arccos[\text{clip}\{-\tan \phi \tan \delta_d, -1, 1\}], \\ p_d &= \text{clip}\{1 - 0.09(16 - \ell_d), 0, 1\}. \end{aligned} \quad (\text{S2})$$

Photoperiod sensitivity depends on the prior-day thermal clock:

$$\begin{aligned} b_d^P &= c_d \mathbf{1}(G_P^{\text{on}} \leq G_{d-1} < G_P^{\text{off}}), \\ u_d &= g_d \{1 - b_d^P + b_d^P p_d\}, \\ U_d &= U_{d-1} + u_d. \end{aligned} \quad (\text{S3})$$

The temperature response for vernalization is

$$v(T) = \begin{cases} 0, & T \leq -4 \text{ or } T \geq 16, \\ (T + 4)/4, & -4 < T < 0, \\ 1, & 0 \leq T \leq 10, \\ (16 - T)/6, & 10 < T < 16. \end{cases} \quad (\text{S4})$$

For vernalization-required cases, define $b_d^V = c_d \mathbf{1}(U_{d-1} \geq U_V^{\text{on}})$, $s_d^V = \mathbf{1}(U_{d-1} < U_V^{\text{off}})$, and $\tilde{V}_d = V_{d-1} + s_d^V v(T_d)$. The candidate and committed vernalization states are

$$\begin{aligned} V_d^* &= \begin{cases} \max\{0, V_{d-1} - 0.5(T_d^{\max} - 30)\}, & \tilde{V}_d < 10 \text{ and } T_d^{\max} > 30, \\ \tilde{V}_d, & \text{otherwise,} \end{cases} \\ V_d &= (1 - b_d^V) V_{d-1} + b_d^V V_d^*, \end{aligned} \quad (\text{S5})$$

$$\begin{aligned} f_d^V &= \begin{cases} 0.3 + 0.7 \min(V_d/40, 1), & b_d^V s_d^V = 1, \\ 1, & \text{otherwise,} \end{cases} \\ A_d &= A_{d-1} + u_d f_d^V. \end{aligned} \quad (\text{S6})$$

Without a vernalization requirement, $V_d = V_{d-1}$, $f_d^V = 1$ and $A_d = U_d$. Photoperiod and vernalization gates use G_{d-1} and U_{d-1} , whereas the vernalization response multiplier uses the updated V_d . Thus current weather affects current development without an observed stage entering the calculation. G , U and A have effective °C d units; V has effective-day units. Effective TPV units are a developmental index and do not establish a measured leaf phyllochron. The retained development multiplier is $q = 1$.

For a supplied developmental threshold C_k , the predicted stage date is

$$\tau_k = \min\{d : c_d = 1, A_d \geq C_k\}. \quad (\text{S7})$$

An unreached stage remains missing. BBCH codes are categorical event labels; numerical BBCH interpolation is absent. Additional thresholds $C_{32}, C_{33}, C_{37}, C_{39}$ are ordered stage-only estimates. The flowering proxy satisfies $C_{65} = C_{51} + 0.2317(C_{85} - C_{51}) = 1165.925558$ effective °C d. The profile tolerance sets for these estimates describe calibration-loss compatibility, rather than confidence intervals.

Final-leaf capacity and juvenile handover

Slots $i = 0, \dots, 6$ denote F1–F7, with F1 the flag leaf; slot $i = 7$ is an aggregate juvenile canopy. Final-leaf visibility and full-unfolding thresholds are

$$\begin{aligned} B_i &= \max\{C_{10}, C_{37} - i\Delta\}, \\ E_i &= B_i + (C_{39} - C_{37}), \end{aligned} \quad (\text{S8})$$

where the selected effective rank spacing is $\Delta = 120$ TPV units. End-day visibility is $c_d \mathbf{1}(A_d \geq B_i)$, and full unfolding is $c_d \mathbf{1}(A_d \geq E_i)$. Availability for disease requires prior-day visibility:

$$\begin{aligned} H_{i,d} &= c_d c_{d-1} \mathbf{1}(A_{d-1} \geq B_i), \\ a_{i,d} &= H_{i,d} \text{clip} \left(\frac{A_d - B_i}{E_i - B_i}, 0, 1 \right), \\ h_{i,d} &= \mathbf{1}(a_{i,d} > 0). \end{aligned} \quad (\text{S9})$$

The initial prior crop indicator is zero. Visibility on the first threshold-crossing day therefore does not immediately enable infection. Daily susceptible renewal associated with final-leaf expansion is

$$\rho_{i,d} = \begin{cases} \max(a_{i,d} - a_{i,d-1}, 0)/a_{i,d}, & a_{i,d} > 0, \\ 0, & a_{i,d} = 0. \end{cases} \quad (\text{S10})$$

The primary juvenile policy gives

$$\begin{aligned} J_d &= \text{clip} \left(\frac{C_{39} - A_d}{C_{39} - C_{31}}, 0, 1 \right), \\ a_{7,d} &= c_d c_{d-1} \mathbf{1}(A_{d-1} \geq C_{10}) J_d, \\ \rho_{7,d} &= \mathbf{1}(a_{7,d} > 0) \{1 - \exp[-\max(T_d, 0)/300]\}. \end{aligned} \quad (\text{S11})$$

The alternative persistent-juvenile scenario sets $J_d = 1$. Juvenile capacity withdrawal is not leaf unfolding and does not activate contact reception. Capacities $a_{i,d}$ are normalized and dimensionless. Natural senescence, absolute green area and true final main-stem leaf number are absent from this capacity formulation. Lower ranks clipped at C_{10} have unresolved shared appearance thresholds.

Normalized tissue, effective sources and daily numerical transitions

Each slot contains susceptible tissue S_i , m latent compartments $L_{i,1}, \dots, L_{i,m}$, visibly affected noninfectious tissue N_i , infectious tissue I_i , and retained damage D_i :

$$S_i + \sum_{r=1}^m L_{i,r} + N_i + I_i + D_i = 1. \quad (\text{S12})$$

All tissue states are normalized fractions. Initial tissue is entirely susceptible. Local source potential has relative unready and competent states Q^u, Q^c , initialized as $((1 - f_0)Q_0, f_0Q_0)$, with $Q_0 = 1$ and $f_0 = 0.5$. These states are aggregate source phenomenology; their units are arbitrary relative-source units, with no measured-spore or reproductive-maturation interpretation.

At the beginning of each included crop day, inactive slots reset to $(1, 0, \dots, 0)$. In active slots, all nonsusceptible compartments are multiplied by $1 - \rho_{i,d}$, and S_i becomes one minus their sum. This operation represents capacity dilution or juvenile renewal. Outside the crop window the complete tissue and source states freeze, with no inactive-slot reset.

One day contains $n = \lceil 1/\Delta t_{\text{req}} \rceil$ equal substeps of length $\eta = 1/n$ d. The frozen setting $\Delta t_{\text{req}} = 0.25$ d gives $n = 4$. Weather and host capacity remain fixed within the day. Define $\theta_d = \max(T_d, 0)/18$, $r_d = 1 - \exp(-P_d/p_*)$, with $p_* = 2$ mm. At the beginning of each substep, area-weighted infectious neighborhood and adjacency are

$$\begin{aligned} Z_{i,d} &= \frac{\sum_{j:h_{j,d}=1} \exp(-|i-j|/\lambda) I_j a_{j,d}}{\sum_{j:h_{j,d}=1} a_{j,d}}, \\ C_{i,d} &= \frac{\sum_{j:h_{j,d}=1, |i-j|=1} I_j a_{j,d}}{\sum_{j:h_{j,d}=1} a_{j,d}}, \end{aligned} \quad (\text{S13})$$

with both quantities zero if the denominator is zero. Here $\lambda = 2$ rank intervals. The four effective establishment hazards are

$$\begin{aligned} z_i^{\text{local}} &= \alpha e_d Q^c \{ (1 - \omega) r_d \exp[-(7-i)/\lambda] + \omega \}, \\ z_i^{\text{imported}} &= \alpha e_d M_d, \\ z_i^{\text{splash}} &= \beta e_d r_d Z_{i,d}, \\ z_i^{\text{contact}} &= \beta e_d \kappa (1 - a_{i,d}) C_{i,d} \mathbf{1}(i \neq 7). \end{aligned} \quad (\text{S14})$$

e_d is dimensionless weather exposure, M_d is supplied relative imported pressure, $\omega = 0.1$, and $\kappa = 0.2$. The selected primary and secondary coefficients are $\alpha = 0.001$ and $\beta = 10$; primary normalization is per relative-source unit per day, and secondary normalization is per day. Their scales depend on the assumed source normalization and tissue capacity. The rank kernel includes the receiving slot; imported pressure does not require local source or precipitation.

For $z_i = \sum_p z_i^p$, newly infected tissue and pathway increments are

$$\begin{aligned} F_i &= S_i \{ 1 - \exp(-\eta z_i) \}, \\ F_i^p &= \begin{cases} F_i z_i^p / z_i, & z_i > 0, \\ 0, & z_i = 0. \end{cases} \end{aligned} \quad (\text{S15})$$

Only active slots receive these updates. Pathway outputs sum substep increments over the day; they are newly affected fractions, without a spore-count or identified causal-attribution interpretation. The thermal transition probabilities are

$$\begin{aligned} q_L &= 1 - \exp(-\eta m \theta_d / \tau_L), \\ q_N &= 1 - \exp(-\eta \theta_d / \tau_N), \\ q_I &= 1 - \exp(-\eta \theta_d / \tau_I). \end{aligned} \quad (\text{S16})$$

With pre-substep states on every right-hand side, the simultaneous update is

$$\begin{aligned} S'_i &= S_i - F_i, \\ L'_{i,1} &= (1 - q_L) L_{i,1} + F_i, \\ L'_{i,r} &= (1 - q_L) L_{i,r} + q_L L_{i,r-1}, \quad r = 2, \dots, m, \\ N'_i &= (1 - q_N) N_i + q_L L_{i,m}, \\ I'_i &= (1 - q_I) I_i + q_N N_i, \\ D'_i &= D_i + q_I I_i. \end{aligned} \quad (\text{S17})$$

New infections do not progress through a latent stage in their substep of entry. Reference durations $\tau_L = 20$, $\tau_N = 3$ and $\tau_I = 21$ d refer to the 18 °C clock; $m = 3$. The latent chain has a continuous-time reference mean of τ_L , while realized residence times follow the explicit finite-substep updates.

For source maturation and common ageing rates $\mu_d = \theta_d e_d / \tau_M$ and $\nu_d = \theta_d / \tau_Q$, with $\tau_M = 30$ and $\tau_Q = 90$ reference d, the implemented source update is

$$\begin{aligned} (Q^u)' &= Q^u \exp[-\eta(\mu_d + \nu_d)], \\ (Q^c)' &= \exp(-\eta\nu_d)\{Q^c + Q^u[1 - \exp(-\eta\mu_d)]\}. \end{aligned} \quad (\text{S18})$$

Consequently $(Q^u + Q^c)' = (Q^u + Q^c) \exp(-\eta\nu_d)$ exactly. Tissue and source proposals use their beginning-of-substep states and are committed together. Nonpositive mean temperature stops the thermal progression and source-ageing clocks; it does not force weather exposure or establishment hazards to zero. Renewal and availability also affect tissue independently of the thermal transition probabilities.

Weather exposure and observation operator

For hourly-bin index $h = 0, \dots, 23$, synthetic hourly temperature is

$$\hat{T}_{d,h} = T_d + \max(T_d^{\max} - T_d, 0) \cos\{2\pi(h + 0.5)/24\}. \quad (\text{S19})$$

Let $s_{d,h} = \exp\{17.625\hat{T}_{d,h}/(243.04 + \hat{T}_{d,h})\}$. A common relative vapor pressure v_d is found by 30 bisection iterations on $[0, \max_h s_{d,h}]$, matching $24^{-1} \sum_h \min(v_d/s_{d,h}, 1) = R_d/100$. Synthetic hourly humidity is $\hat{R}_{d,h} = 100 \min(v_d/s_{d,h}, 1)$, with exact zero and 100% values retained at those daily-mean boundaries. Then

$$\begin{aligned} H_{d,h} &= \mathbf{1}(\hat{R}_{d,h} \geq 90 - 10^{-7}), \\ f_d^R &= 1 - \exp[-P_d/(24r_*)], \end{aligned} \quad (\text{S20})$$

$$e_d = \frac{1}{24} \sum_{h=0}^{23} \exp\left[-\frac{1}{2} \left(\frac{\hat{T}_{d,h} - 18}{8}\right)^2\right] \{H_{d,h} + (1 - H_{d,h})f_d^R\}. \quad (\text{S21})$$

The selected effective precipitation intensity is $r_* = 0.45 \text{ mm h}^{-1}$. It is a meteorology-only training estimate of median daily precipitation divided by observed positive-rain hours on eligible days with mean temperature above 2 °C. Synthetic humidity assumes constant vapor pressure, and the rain overlap term assumes independence of rain timing and humid bins. These are daily exposure approximations, without observed hourly humidity or leaf-wetness reconstruction. An explicit supplied exposure in $[0, 1]$ overrides this operator. Missing imported pressure inherits $M_d = 0.1$; an explicit zero remains zero.

End-day latent, symptomatic and affected fractions are

$$\begin{aligned} L_{i,d}^{\text{total}} &= \sum_r L_{i,r,d}, \\ X_{i,d} &= N_{i,d} + I_{i,d} + D_{i,d}, \\ B_{i,d}^{\text{affected}} &= L_{i,d}^{\text{total}} + X_{i,d} = 1 - S_{i,d}. \end{aligned} \quad (\text{S22})$$

The modeled infection-event date is the first included day with $B_{i,d}^{\text{affected}} \geq \epsilon$, and the symptom-event date is the first included day with $X_{i,d} \geq \epsilon$, where $\epsilon = 0.001$. This is a fixed detection operator, rather than a biological infection threshold. Observed positive magnitudes are reduced to zero/positive signs for fitting. A preceding zero at day l and first positive at day u imply an onset interval $(l, u]$; a missing preceding zero gives left censoring, and an all-zero history gives right censoring. Later zero returns are persistence violations and do not replace the first-positive interval.

For a finite modeled symptom day t , interval-excess loss is

$$\begin{aligned} \mathcal{D}(t; l, u) = & \mathbf{1}(l \text{ finite}) \max(l + 1 - t, 0) \\ & + \mathbf{1}(u \text{ finite}) \max(t - u, 0). \end{aligned} \quad (\text{S23})$$

A missing modeled event with an observed positive receives $\max(30, d_{\text{end}} + 1 - u)$ d loss; a missing event in an all-zero history is compatible through the final assessment. The observation operator targets first detectable modeled symptoms. It does not identify observed infection dates or fit disease-percentage magnitude. For assessment or interval rows, the hierarchy weight is $[K n_f(k) n_\ell(k, f) n_r(k, f, \ell)]^{-1}$, where K is the number of coordinate-year groups, n_f their field counts, n_ℓ the represented leaf counts within a field, and n_r the rows within a leaf. Disease selection uses three inner location folds of the original 28 fields, a 144-setting grid, pooled top-three two-sided interval loss, balanced assessment-error tie breaking and fixed enumeration. The selected secondary coefficient lies at the tested upper bound. Effective coefficients remain conditional on the fixed source normalization, daily step and ecological assumptions; the fit does not identify separate biological source contributions.

Conditional healthy-area-duration diagnostic

For F1–F3, an explicitly supplied nominal maximum upper-three-leaf reference LAI L_* defines the scenario area

$$\mathcal{A}_{i,d} = L_* a_{i,d} / 3, \quad i = 0, 1, 2. \quad (\text{S24})$$

$\mathcal{A}$ has units $\text{m}^2 \text{ lamina m}^{-2} \text{ ground}$; $L_* = 1$ in the standardized scenario. The public daily runtime and field full-season replay use $f_{i,d} = X_{i,d}$ as the assumed functional-loss fraction. The published climate replay applies a detection gate, $f_{i,d} = X_{i,d} \mathbf{1}(\tau_i^{\text{symptom}} \text{ exists and } d \geq \tau_i^{\text{symptom}})$. Climate subthreshold damage therefore contributes zero before the first detected symptom, whereas the field replay and public runtime retain that subthreshold contribution. Latent tissue is excluded in both definitions. Neither mapping is a measured functional-green-area conversion.

With the inclusive flowering-proxy–BBCH85 window $\mathcal{W} = \{d : \tau_{65} \leq d \leq \tau_{85}\}$, daily rectangular integration gives

$$\begin{aligned} \text{HAD}_3^{\text{ref}} &= \sum_{d \in \mathcal{W}} \sum_{i=0}^2 \mathcal{A}_{i,d} (1 \text{ d}), \\ \text{HAD}_3^{\text{lost}} &= \sum_{d \in \mathcal{W}} \sum_{i=0}^2 \mathcal{A}_{i,d} f_{i,d} (1 \text{ d}), \end{aligned} \quad (\text{S25})$$

$$\begin{aligned} \text{HAD}_3^{\text{healthy}} &= \text{HAD}_3^{\text{ref}} - \text{HAD}_3^{\text{lost}}, \\ \Delta Y_b &= b \text{HAD}_3^{\text{lost}}. \end{aligned} \quad (\text{S26})$$

HAD units are $\text{m}^2 \text{ green lamina m}^{-2} \text{ ground d}$, equivalent to GLAI d. The slope scenarios $b = 0.0141, 0.0180, 0.0207$ have units t ha^{-1} per GLAI d. Their products have units t ha^{-1} , conditional on the nominal reference canopy and conversion; the unit-LAI scenario scales linearly with L_* . The source coefficient window began at GS31, with a different integration convention and unrecovered exact terminal rule; the flowering–85 diagnostic is a restricted-window adaptation. A separate climate GS31–85 diagnostic uses the same definitions with start date τ_{31} . Slope ranges are cultivar scenarios, rather than confidence intervals. Incomplete windows yield missing conditional losses. Percentage loss is $100\Delta Y_b / Y_{\text{ref}}$ only when an independent positive reference yield Y_{ref} is supplied. No yield is inferred from nominal LAI, and losses above an independent reference yield are flagged without clipping. An external-area runtime mode integrates caller-supplied reference LAI and functional-loss fractions instead. Actual field-yield forecasts remain undefined.

Spatial paired estimation and Monte Carlo sampling error

The current climate replay uses 16 geographical strata, four independent draws with replacement per stratum, and selection probability proportional to registered harvested area within each stratum. The 64 draw identities comprise 62 unique cells; coincident cells retain separate draw identities. If W_h is eligible area in stratum h divided by total eligible area, each of its $n_h = 4$ draws has weight $w_{hr} = W_h/n_h$. Missing calendars are excluded from the eligible population; missing weather, unreached BBCH85 and unsupported temperatures yield missing outcomes. Calendar type and actual wheat genetics remain scenario assumptions.

For climate model g , SSP s , period p , draw r , and relative year $t = 1, \dots, 30$, let y_{gsprt} denote a metric and v_{gsprt} indicate a finite metric. Define

$$\begin{aligned} x_{gspr} &= \frac{1}{30} \sum_t v_{gsprt} y_{gsprt}, \\ z_{gspr} &= \frac{1}{30} \sum_t v_{gsprt}, \\ \hat{c}_{gsp} &= \sum_r w_r z_{gspr}, \\ \hat{\mu}_{gsp} &= \frac{\sum_r w_r x_{gspr}}{\hat{c}_{gsp}}. \end{aligned} \tag{S27}$$

The numerator convention assigns zero contribution to missing values, including when all 30 values for a draw are missing; $\hat{c}$ is metric-specific area-time coverage. Period estimates are ratios of equal-year area-time totals. They are not averages of separately normalized annual means. Event-date metrics condition on detected events because absent events remain missing.

Historical and future outcomes are paired within the same draw, climate model and SSP at the same relative-year position. Let a_{rt} and b_{rt} be historical and future metrics, and $v_{rt}^\cap = \mathbf{1}(a_{rt} \text{ and } b_{rt} \text{ finite})$. Then

$$\begin{aligned} x_r^\Delta &= \frac{1}{30} \sum_t v_{rt}^\cap (b_{rt} - a_{rt}), \\ z_r^\cap &= \frac{1}{30} \sum_t v_{rt}^\cap, \\ \hat{\Delta} &= \frac{\sum_r w_r x_r^\Delta}{\sum_r w_r z_r^\cap}. \end{aligned} \tag{S28}$$

Historical and future means on this same common support satisfy $\hat{\Delta} = \hat{\mu}_b^\cap - \hat{\mu}_a^\cap$. Pairing aligns positions in 30-year periods, without asserting a physical correspondence between individual historical and future years.

For any ratio estimate $\hat{\mu} = \sum_r w_r x_r / \hat{c}$, let $\psi_r = (x_r - \hat{\mu} z_r) / \hat{c}$. The spatial Monte Carlo standard error is

$$\begin{aligned} \widehat{\text{MCSE}}(\hat{\mu}) &= \left[\sum_h \frac{W_h^2}{n_h} s_h^2(\psi) \right]^{1/2}, \\ s_h^2(\psi) &= \frac{1}{n_h - 1} \sum_{r \in h} (\psi_r - \bar{\psi}_h)^2. \end{aligned} \tag{S29}$$

The same expression uses $x_r^\Delta, z_r^\cap$ for paired changes. No finite-population correction applies to independent replacement draws. The three-model ensemble mean is the mean of model-specific ratios. Because every climate model shares spatial draws, its MCSE uses $\bar{\psi}_r = 3^{-1} \sum_g \psi_{gr}$ inside Equation S29, retaining cross-model spatial covariance. Model minima and maxima describe three-model spread, rather than confidence intervals. Spatial MCSE quantifies simulation-sampling error conditional on the registered forcing, fit and scenarios; it does not include parameter, observation, source-process, calendar or

climate-projection uncertainty. The archived severity-model parameter ensembles and full-grid anchors have a different model identity and do not supply uncertainty intervals for the current source/tissue model.

Model identity and evidence scope

The current runtime is model.wheat_stb version 0.1.0, composed with the source/tissue kernel in model/seasonal_septoria/overwinter.py, stage-anchored canopy in leaf_phenology.py, and daily exposure operator in wetness.py. The current selected disease fit is overwinter_source_model/frozen_selected_fit.json; the ordered stage fit and retained donor TPV record are separate explicit inputs. The earlier seasonal_calibration_v1 severity formulation and infection_and_upper3_yield_v1_20261006 event formulation retain their original identities and archived outcomes. Only the stage-calibrated flowering threshold is inherited from the latter. Source hashes, configuration hashes and the parameter inventory identify exact implementations and input records. Numerical consistency of the equations establishes computational correspondence, without establishing predictive accuracy, measured source identification, observed infection dates or actual yield loss.